Holistic hamstring health: not just the Nordic hamstring exercise

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INTRODUCTION

Hamstring strain injuries (HSIs) are the most prevalent in team sports, accounting for 12%–26% of injuries in Australian rules football, American football, football, rugby and track and field.[1] The biceps femoris is the most commonly injured muscle with 53%–68% of injuries occurring during sprinting.[1] In European football, the incidence and recurrence of HSI has continued to rise, while in Australian rules football, a notable reduction in HSI recurrences has been reported.[1] It is possible that evidence-based hamstring injury prevention is not adopted or adhered to in some elite level football teams,[2] which may explain the rise in HSI.

Although not an exhaustive list, HSI risk factors include: age, previous injury, strength imbalance, flexibility, fatigue[1] and low eccentric strength.[3] There is a growing body of evidence on the Nordic hamstring exercise (NHE) and its impact on HSI reduction.[4] However, there may be misconceptions (fuelled by social media) that this is the only exercise used to prevent HSI. The NHE is not the 'silver bullet' that critics imply practitioners claim it to be, and it is unlikely that experienced practitioners solely rely on the NHE when dealing with HSI.

NORDIC HAMSTRING EXERCISE

The popularity of the NHE may lie in its ease of use, requiring no equipment and being effective at reducing the risk of HSI.[4] It is possible that the intervention was adopted out of necessity to address HSIs in sports like football[4] that may have only recently embraced more traditional strength training. While seen by some to be non-functional, the NHE combined with sports-specific training in professional and amateur football players has been shown to reduce both the incidence and recurrence rate of HSI by 60% and 85%, respectively, following a 10-week intervention programme.[4] Greater eccentric hamstring strength may also offset the likelihood of injury in older or previously injured athletes.[3] This appears to be an effective intervention, requiring minimal effort when compliance is adequate.

The invention of devices like the 'Nord-Bord' allows for quick, easy and reliable measurements of eccentric knee flexor strength, which may explain the recent popularity of the NHE. These devices may provide a link between testing and training where daily feedback is received and actionable data are generated. However, the NHE may be limited in its effectiveness by only training eccentric knee flexor strength in a knee dominant action. A low adoption rate and poor adherence to the exercise in elite level football[2] may be due to the initial soreness experienced by some with this exercise.[4]

HAMSTRING HEALTH: TWO MISSING PARTS OF THE PUZZLE

While the NHE is an effective prevention tool, we continue to observe a high incidence of HSI. A greater appreciation for the multifactorial nature of HSI while also addressing the primary injury mechanism—sprinting[1] is required. We propose a more holistic approach to hamstring health (figure 1).

Exercise selection

During the late swing phase of running, the biceps femoris and other hamstring muscles function eccentrically to resist hip flexion and decelerate knee extension,[5] where they undergo large ranges of motion and different activation patterns. It would therefore be advisable to train the hamstrings with both hip and knee dominant exercises.

Holistic hamstring health (figure 1) suggests alternative options including the use of other hip and knee dominant exercises reported in elite level football.[2] A growing body of evidence exists for the 'hip extension exercise'. It has been shown to target the biceps femoris long head and semitendinosus while the NHE preferentially recruits the semitendinosus but also elicits the greatest absolute activation of biceps femoris compared with other hamstring exercises.[6] Overloading the eccentric contractions may elongate muscle fascicles which could mediate the protective mechanism of improving muscle strength at longer lengths.[6]

High-speed running (sprinting)

Exposing athletes to increased weekly sprint distances (90–120 m above 95% maximum velocity) and exposures (6–10 efforts) in team-based running sports has also been shown to have a protective effect on lower limb injuries.[7] Importantly, the acute high-speed running load should be progressed gradually, avoiding large sudden increases which may increase the odds of HSI.[8]

CONCLUSION

The NHE is only one method of strengthening, which is important for hamstring health. We would advocate that both hip and knee dominant exercises be included in an injury prevention programme. While the exercises discussed develop eccentric hamstring strength, the contractions occur at a much slower rate than during sprinting and it would be logical to regularly and

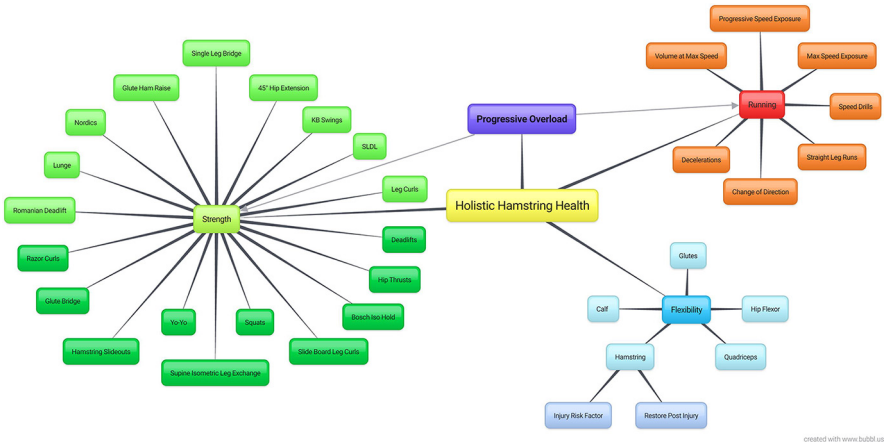

Figure 1 Holistic hamstring health: a holistic approach to hamstring strain injury prevention and hamstring strengthening.

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progressively expose the athlete to high-speed running. Only when all these factors are addressed in a multifaceted approach can we hope to achieve holistic hamstring health.

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